

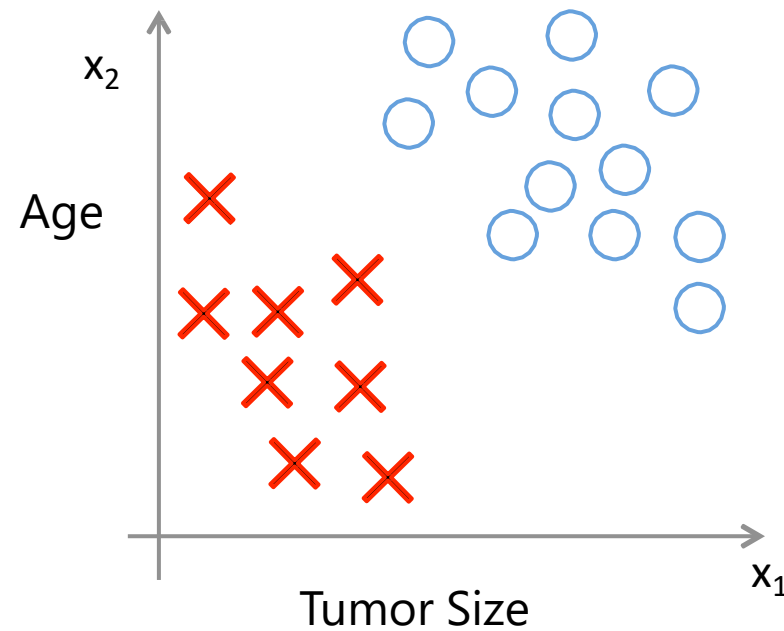
Support Vector Machines

Prof. Giuseppe Serra

Università degli Studi di Udine

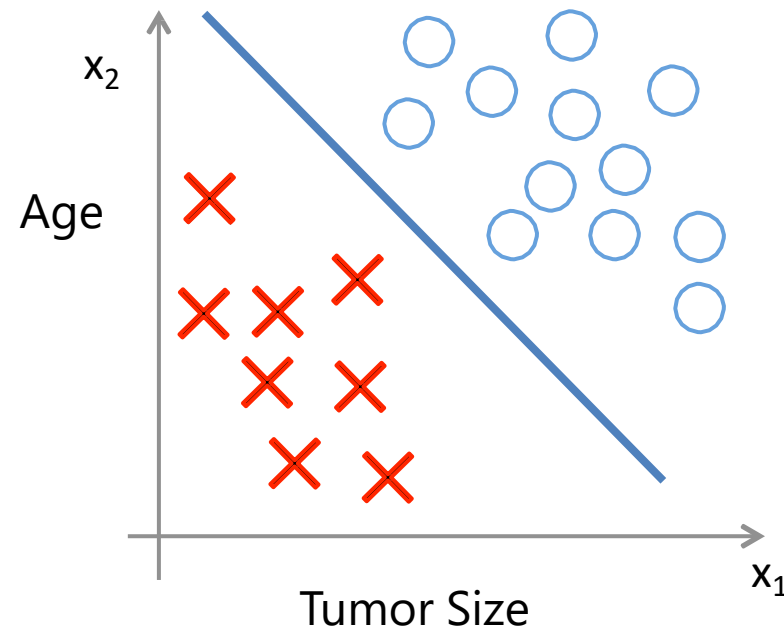
Classification

Suppose we have a tumor classification problem and we have as features both the age of the patients and the tumour sizes. The dataset could look like this:



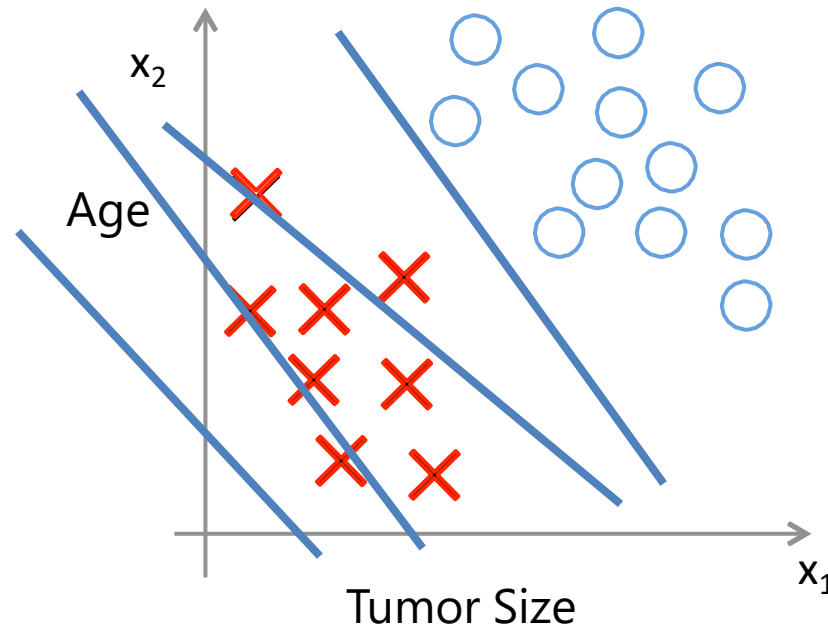
Classification

A learning algorithm can separate out the two classes. Linear Support Vector Machines tries to estimate the decision boundary (i.e. the line that separates the data)



Classification

In 2D space the decision boundary is modelled by three parameters: θ_0 , θ_1 and θ_2 . Decision boundary $\theta_0 + \theta_1 x_1 + \theta_2 x_2 = 0$

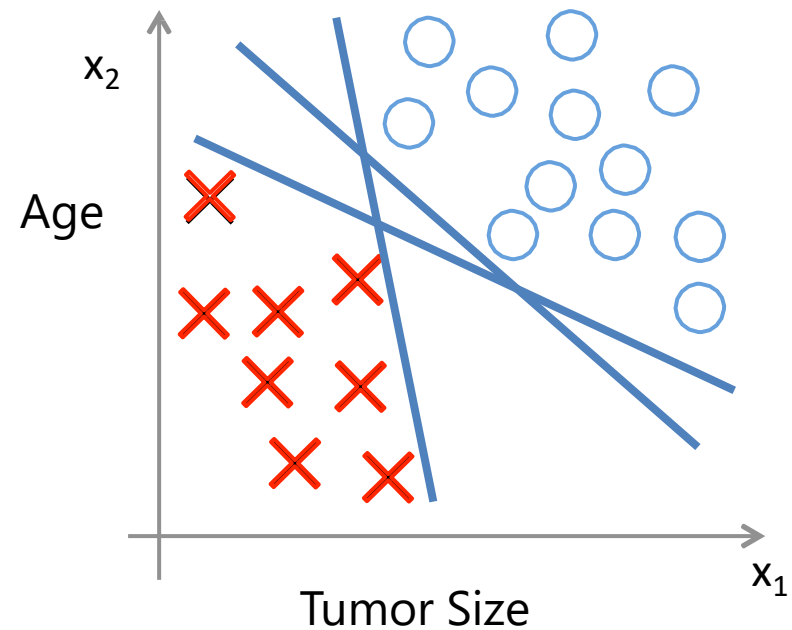


Given θ_0 , θ_1 , θ_2 , a Cost Function can be computed on the training set. Using a minimization algorithm the system can adapt the parameters θ_0 , θ_1 , θ_2 (i.e. the decision boundary).

Classification

Goal: minimize θ_0 , θ_1 and θ_2 so that the decision boundary separates the positive and the negative samples.

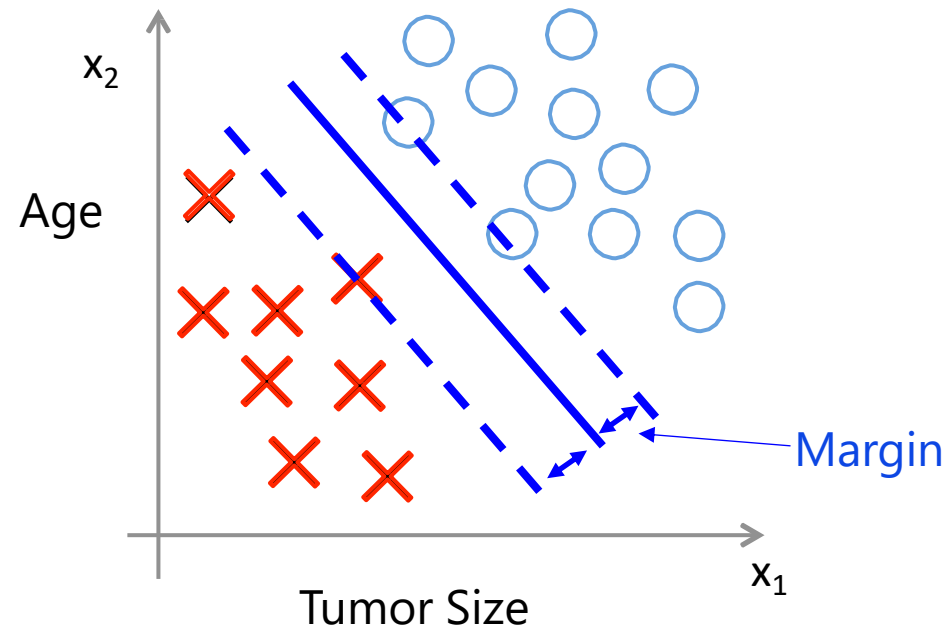
Even in this case we have multiple many possibilities



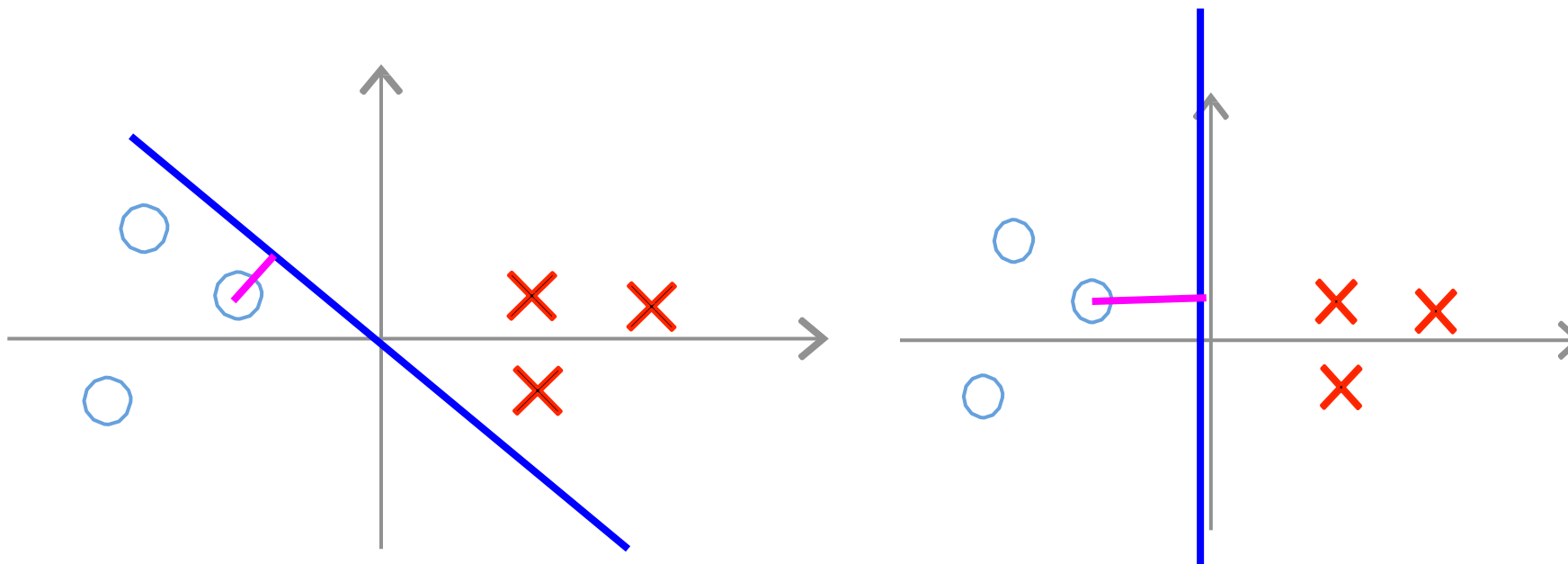
Support Vector Machine

Support Vector Machine (SVM) Goal: minimize θ_0 , θ_1 and θ_2 so that the decision boundary separates the positive and the negative samples with the "large margin".

So SVM chooses the decision boundary so that the distance from it to the nearest data point on each side is maximized.



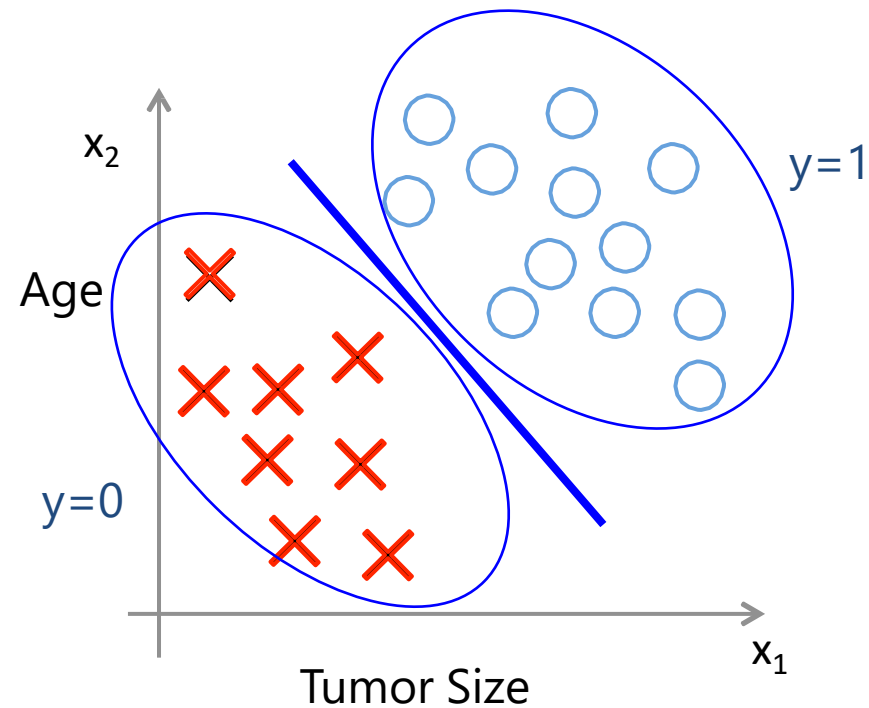
Cost Function



Decision Boundary

Given a new test sample $\hat{x} = (\hat{x}_1, \hat{x}_2)$ and a decision boundary $\theta_0 + \theta_1 x_1 + \theta_2 x_2$

The classifier predicts " $y = 1$ " if $\theta_0 + \theta_1 \hat{x}_1 + \theta_2 \hat{x}_2 \geq 0$



Logistic Regression with linear decision boundaries

Let's consider a 2D case $h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$

with $\theta = [\theta_0, \theta_1, \theta_2]^T, x = [1, x_1, x_2]^T$

Using the Logic Regression Algorithm we can obtain

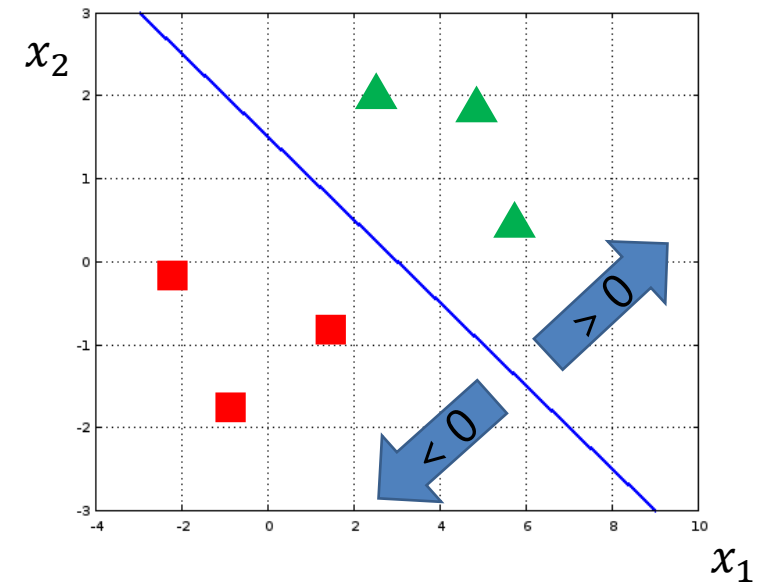
$$\theta = [-3, 1, 2]$$

Remember:

- $h_{\theta}(x) = g(\theta^T x) > 0.5$ when $\theta^T x > 0$

Then the decision boundary is

$$h_{\theta}(x) = 0.5 \Rightarrow \theta^T x = 0 \Rightarrow -3 + x_1 + 2x_2 = 0$$



Decision boundary ($\theta^T x = 0$)

Decision boundary

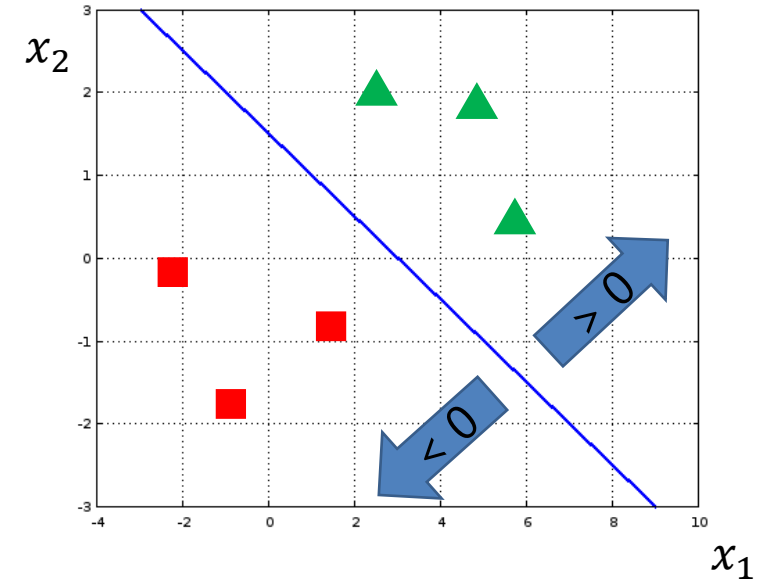
The decision boundary is:

$$-3 + x_1 + 2x_2 = 0$$

$$\begin{matrix} [1 & 2] \\ \mathbf{w}^T & \mathbf{x} \end{matrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - 3 = 0$$

In general an hyperplane can be written as following:

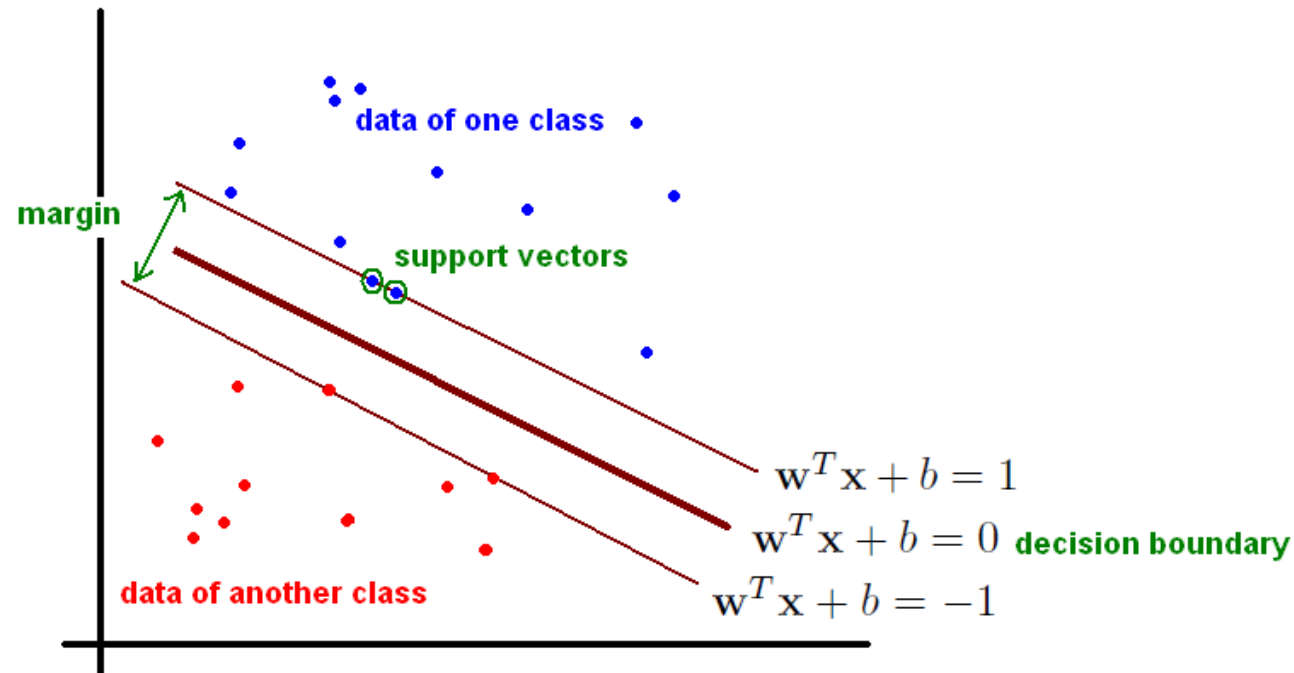
$$\mathbf{w}^T \mathbf{x} + b = 0$$



Initial Assumption: Data are Linearly separable

We assume the data is linearly separable

For a more mathematical expression, it's preferable to renormalize our dataset so that support vectors will lie on the margin.



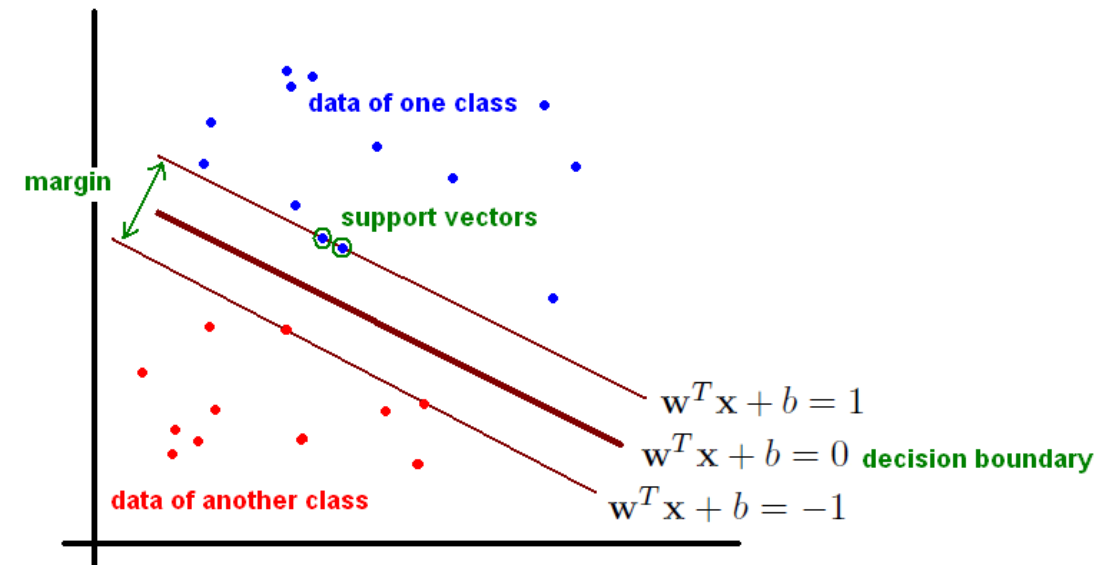
SVM

Implementation a SVM is boiled down to selecting the variable $\mathbf{w}$ and b so that our training data can be described by:

- $\mathbf{w}^T x_i + b \geq 1$ for $y_i = +1$
- $\mathbf{w}^T x_i + b \leq -1$ for $y_i = -1$

These equations can be combined into:

- $y_i(\mathbf{w}^T x_i + b) \geq 1$ for $\forall x_i$

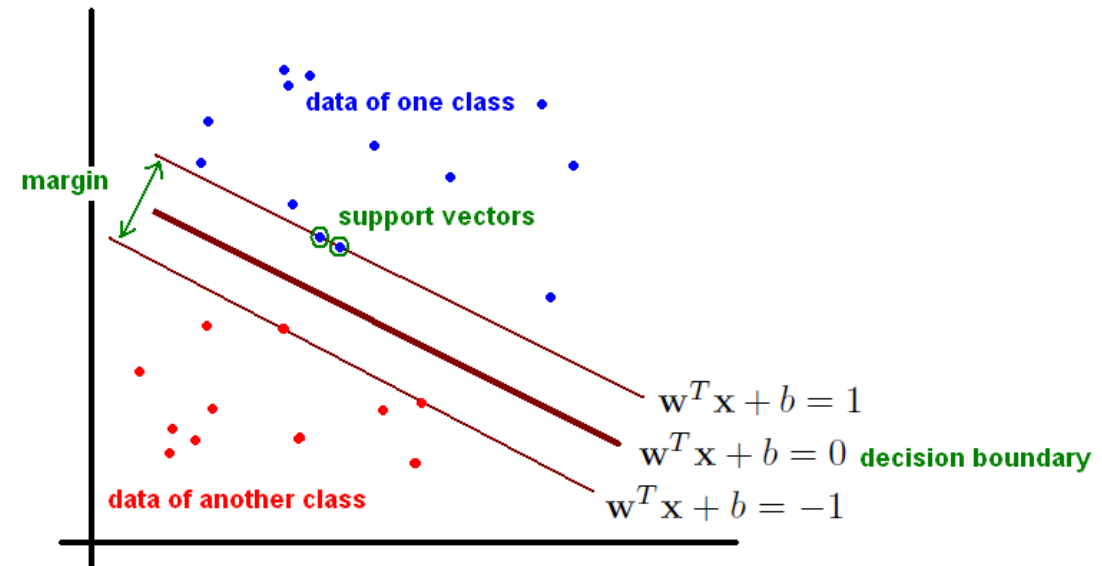


Margin

We can define two boundaries

$$\mathbf{w}^T \mathbf{x} + b = 1$$

$$\mathbf{w}^T \mathbf{x} + b = -1$$



What is the distance between these newly added boundaries?

Perpendicular vector of the Decision Boundary

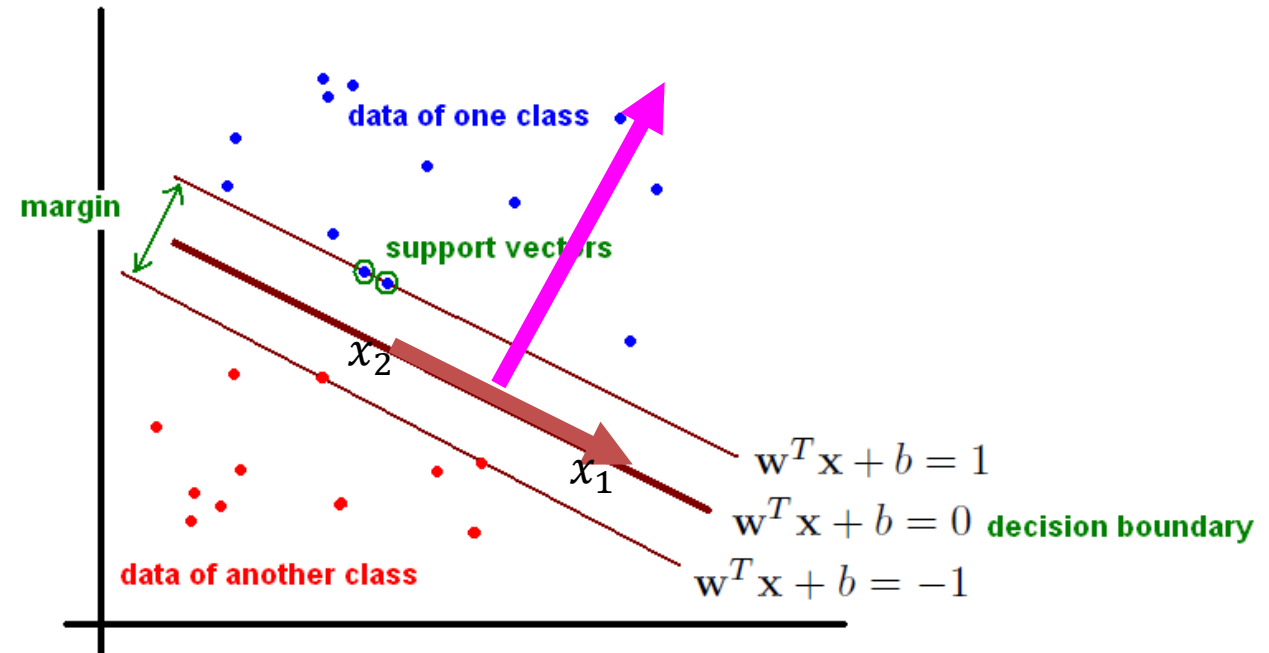
Pick two arbitrary points x_1 and x_2 to lie on $w^T x + b = 0$

$$w^T x_1 + b = 0$$

$$w^T x_2 + b = 0$$

then

$$w^T (x_1 - x_2) = 0$$



Margin

The two boundaries are parallel

Pick an arbitrary point $\mathbf{x}_1$ to lie on $\mathbf{w}^T \mathbf{x} + b = -1$

The close point on $\mathbf{w}^T \mathbf{x} + b = 1$ is $\mathbf{x}_2 = \mathbf{x}_1 + \lambda \mathbf{w}$ (recall $\mathbf{w} \perp$ both lines)

$$\mathbf{w}^T \mathbf{x}_2 + b = 1 \text{ where } \mathbf{x}_2 = \mathbf{x}_1 + \lambda \mathbf{w}$$

$$\mathbf{w}^T (\mathbf{x}_1 + \lambda \mathbf{w}) + b = 1$$

$$\mathbf{w}^T \mathbf{x}_1 + b + \lambda \mathbf{w}^T \mathbf{w} = 1 \text{ where } \mathbf{w}^T \mathbf{x}_1 + b = -1$$

$$-1 + \lambda \mathbf{w}^T \mathbf{w} = 1$$

$$\lambda \mathbf{w}^T \mathbf{w} = 2$$

$$\lambda = \frac{2}{\mathbf{w}^T \mathbf{w}} = \frac{2}{\|\mathbf{w}\|^2}$$

$$\text{Thus, the distance between } \mathbf{x}_2 \text{ and } \mathbf{x}_1 \text{ is } \lambda \|\mathbf{w}\| = \frac{2}{\|\mathbf{w}\|^2} \|\mathbf{w}\| = \frac{2}{\|\mathbf{w}\|} = \frac{2}{\sqrt{\mathbf{w}^T \mathbf{w}}}$$

Margin

And so the distance $\lambda \mathbf{w}$ is $\frac{2}{\|\mathbf{w}\|^2} \|\mathbf{w}\| = \frac{2}{\|\mathbf{w}\|} = \frac{2}{\sqrt{\mathbf{w}^T \mathbf{w}}}$

Thus, we want to maximize $\frac{2}{\sqrt{\mathbf{w}^T \mathbf{w}}}$

Which is equivalent to minimizing $\frac{\sqrt{\mathbf{w}^T \mathbf{w}}}{2}$

Which is in turn equivalent in $\frac{\mathbf{w}^T \mathbf{w}}{2}$

SVM Cost Function

The SVM cost Function for linearly-separable data is:

$$\min_{w,b} \frac{w^T w}{2}$$

Subject to $y_i(w^T x_i + b) \geq 1$ for $\forall x_i$

Soft-margin Extension (no separable data)

We can introduce slack variables – an $\epsilon_i > 0$ for each x_i . In this case, the cost Function is:

$$\min_{w,b} \frac{w^T w}{2} + C \sum_i \epsilon_i$$

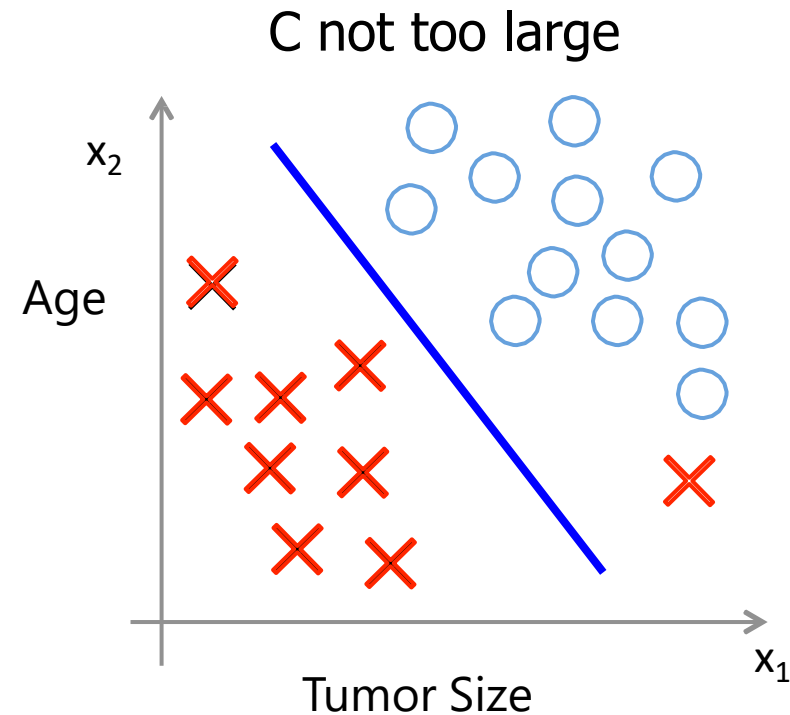
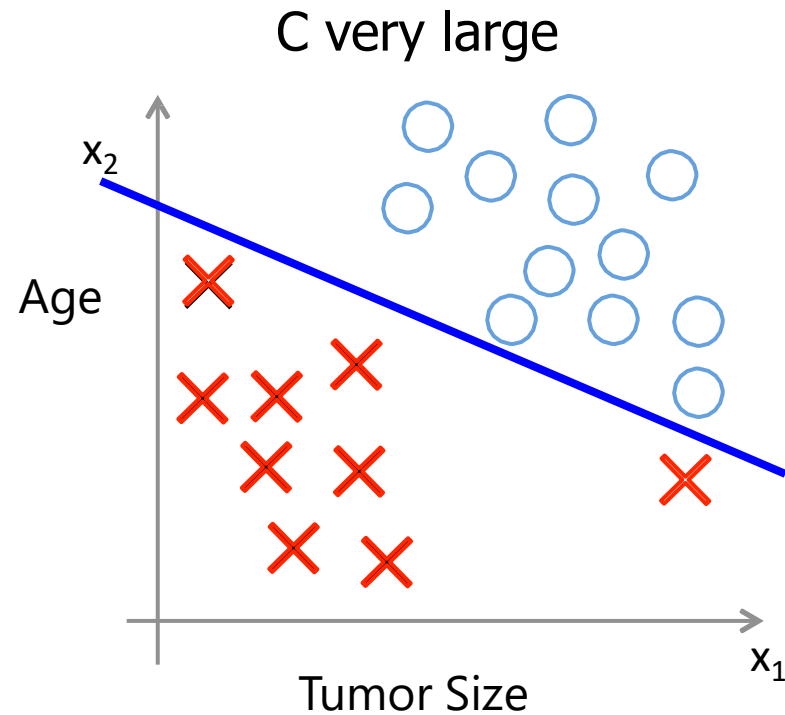
Subject to $y_i(w^T x_i + b) \geq 1 - \epsilon_i$ and $\epsilon_i > 0$ for $\forall x_i$

The two Cost Functions can be minimized as Quadratic Programming Problem.

Regularization term

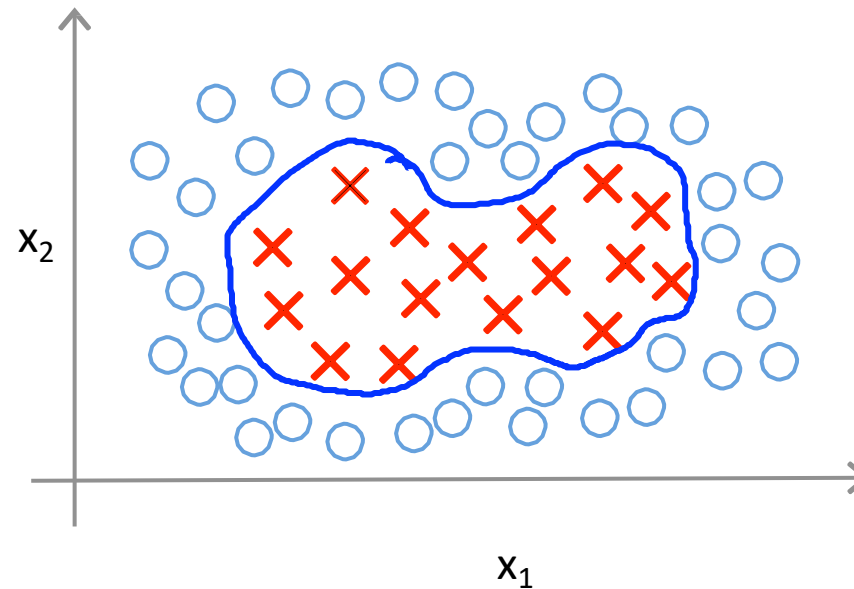
The C parameter in SVM is a positive value that controls the penalty for misclassified training examples.

A Large C parameters tells the SVM to try to classify all the examples correctly



Support Vector Machines with kernels

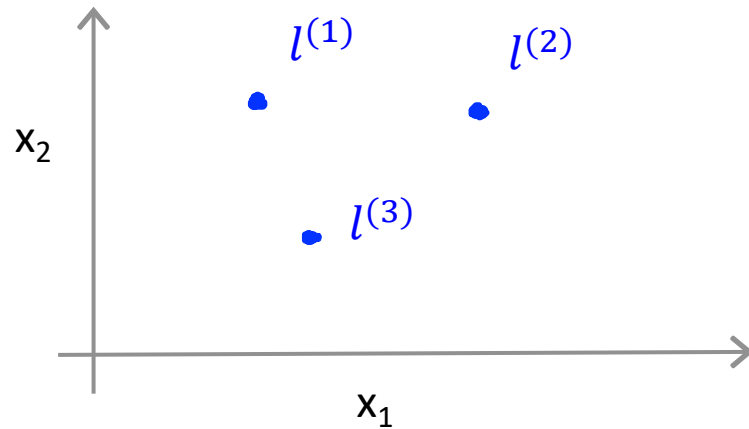
Support Vector Machines with kernels can compute a non-linear decision boundary



We can use a polynomial decision hypothesis ... but we have to decide the degree of the polynomial.

Gaussian Kernels

Gaussian Kernels



$$f_1 = \text{similarity}(x, l^{(1)}) = \exp\left(-\frac{\|x - l^{(1)}\|^2}{2\sigma^2}\right)$$

$$f_2 = \text{similarity}(x, l^{(2)}) = \exp\left(-\frac{\|x - l^{(2)}\|^2}{2\sigma^2}\right)$$

$$f_3 = \text{similarity}(x, l^{(3)}) = \exp\left(-\frac{\|x - l^{(3)}\|^2}{2\sigma^2}\right)$$

Given x , compute new feature depending on proximity to landmarks $l^{(1)}, l^{(2)}, l^{(3)}$

if $x \approx l^{(1)} \rightarrow f_1 \approx 1; f_2 \approx 0; f_3 \approx 0;$

So, these features measure how similar x is from one of your landmarks

Kernels and Similarity (details)

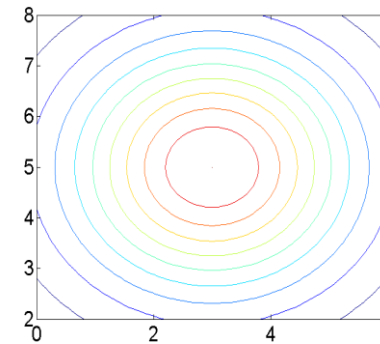
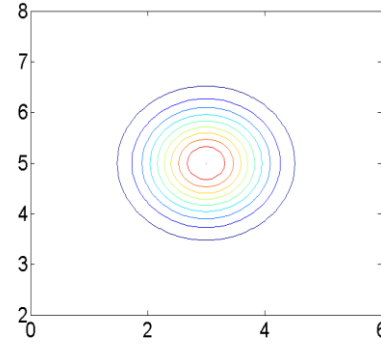
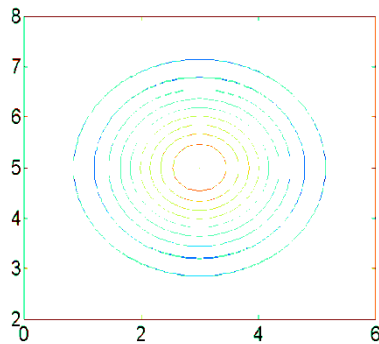
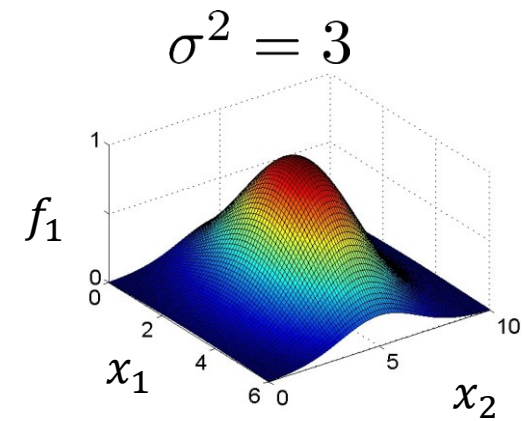
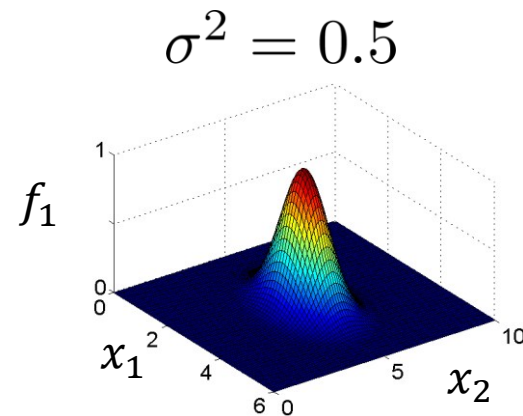
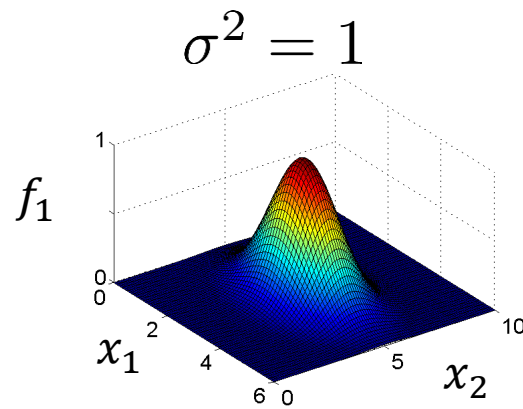
$$f_1 = \text{similarity}(x, l^{(1)}) = \exp\left(-\frac{\|x - l^{(1)}\|^2}{2\sigma^2}\right) = \exp\left(-\frac{\sum_{j=1}^n (x_j - l_j^{(1)})^2}{2\sigma^2}\right)$$

$$\text{if } x \approx l^{(1)} \rightarrow f_1 \approx \exp\left(-\frac{0^2}{2\sigma^2}\right) \approx 1$$

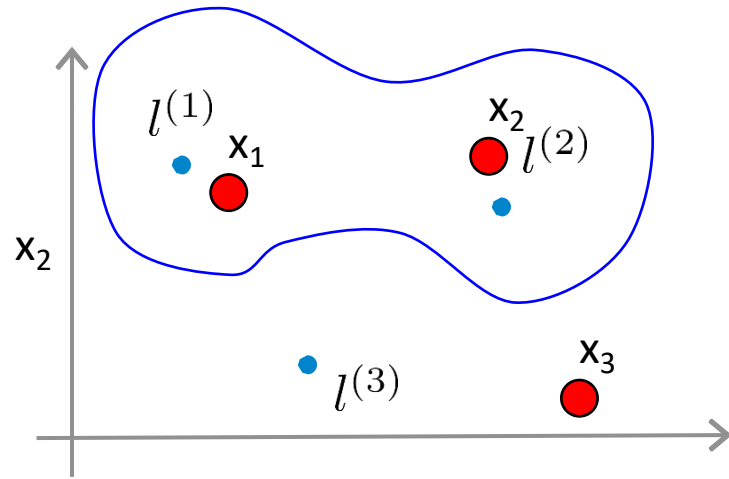
$$\text{if } x \text{ is far from } l^{(1)} \rightarrow f_1 \approx \exp\left(-\frac{(\text{large number})^2}{2\sigma^2}\right) \approx 0$$

Radius of the Gaussian

$$l^{(1)} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}, \quad f_1 = \exp\left(-\frac{\|x - l^{(1)}\|^2}{2\sigma^2}\right)$$



Feature Space



With Kernels the decision boundary is based on these new features f_1, f_2 and f_3 :

$$\theta_0 + \theta_1 f_1 + \theta_2 f_2 + \theta_3 f_3 \geq 0$$

Suppose a Kernelized SVMs obtain the following value:

$$\theta_0 = -0.5; \theta_1 = 1; \theta_2 = 1; \theta_3 = 0$$

For these three points we have:

$$x_1 \rightarrow f_1 \approx 1; f_2 \approx 0; f_3 \approx 0 \rightarrow -0.5 + 1 * 1 + 1 * 0 + 0 * 0 \geq 0 \rightarrow -0.5 + 1 = 0.5 \geq 0 \rightarrow y = 1$$

$$x_2 \rightarrow f_1 \approx 0; f_2 \approx 1; f_3 \approx 0 \rightarrow -0.5 + 1 \geq 0 = 0.5 \geq 0 \rightarrow y = 1$$

$$x_3 \rightarrow f_1 \approx 0; f_2 \approx 0; f_3 \approx 0 \rightarrow -0.5 < 0 \rightarrow y = 0$$

Choosing the landmarks

Given $(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})$,

choose $l^{(1)} = x^{(1)}, l^{(2)} = x^{(2)}, \dots, l^{(m)} = x^{(m)}$.

Landmarks are located exactly in the same locations as the training examples.

Given example $x^{(i)} \in R^{n+1}$ is mapped in $f^{(i)} \in R^m$:

$$f_1^{(i)} = \text{similarity}(x^{(i)}, l^{(1)})$$

$$f_2^{(i)} = \text{similarity}(x^{(i)}, l^{(2)})$$

$$\vdots$$

$$f_m^{(i)} = \text{similarity}(x^{(i)}, l^{(m)})$$

My new features measure how close an example is to one of element of the training set

SVM software package

Use SVM software package (e.g. scikit-learn, liblinear ...) to find parameters θ

Need to specify

- Choice of parameter C
- Choice of kernel function
 - No kernel ("Linear Kernel") if your feature vector lies in high dimension
 - Gaussian kernel (small number of features). You need to choose σ . In some implementation (for example in scikit-learn) the Gaussian Kernel is defined as:
 - $similarity(x, l^{(1)}) = \exp(-\gamma \|x - l^{(1)}\|^2)$ instead of $similarity(x, l^{(1)}) = \exp(-\frac{\|x - l^{(1)}\|^2}{2\sigma^2})$
 - So, gamma is the parameter to choose